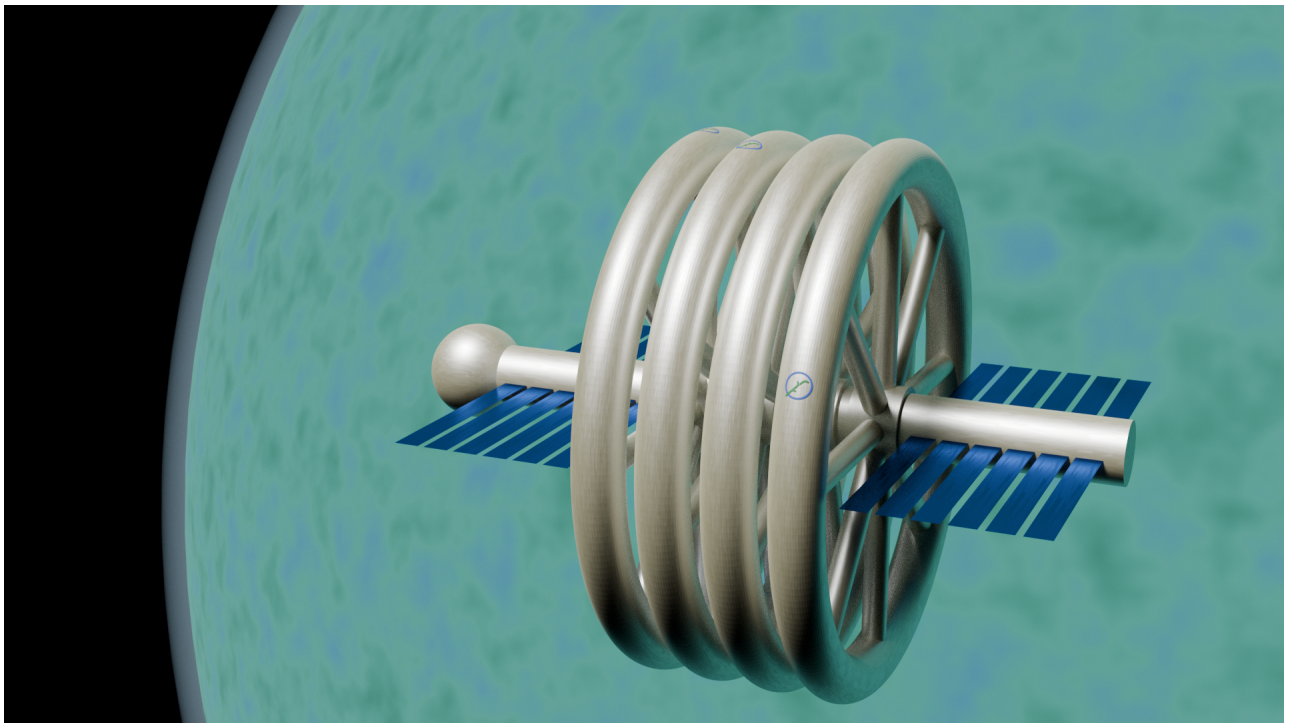




SPROUT

Conceptual Design Study



prepared by

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Abstract (151 words). Since the very dawn of time, humankind has dreamed to “[slip] the surly bonds of earth.” It was not until the latter half of the 20th century, however, that our species first achieved crewed spaceflight and set foot on another celestial body. Contemporary advancements in the sciences of materials, thermodynamics, and biology have prepared humanity uniquely to “scale up life in space” in the 21st century. Before human life in space can be sustainable, we must validate methods of cultivating calorically and nutritionally dense crops, sufficient for sustaining a crewed orbiter, in a space environment. Hence, we propose the Strategically Positioned Rotating Orbital station to facilitate Understanding and Transition of space agriculture (SPROUT) mission. The SPROUT orbiter provides an artificial gravity environment for identifying optimal cereal grains and powerhouse crops for extended mission sustainment, as well as transitioning space agriculture from small-scale experiments to the new era of space urbanism.

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Chapter 1

Introduction

1.1 Project Description

1.1.1 Background

Humankind has dreamed to “[slip] the surly bonds of earth” since the very dawn of time [6]. It was not until the 20th century—mere decades ago—that our species first achieved crewed spaceflight, and shortly after the first crewed missions to Earth orbit came the first crewed missions to the lunar surface. Today, launch cost to LEO per kilogram is decreasing at a pace never before imagined due to stronger and lighter materials, efficient propulsion systems, reusable boosters, ridesharing, and countless other wondrous technologies.

The sparks of such rapid transformation have caught the tinder of imagination in innumerable scientists and engineers, as well as the global public at large. What will our future relationship to space look like? Will humans find ways to live in space sustainably? Will the benefits of spaceflight technology become increasingly accessible to all people, or will it remain in the hands of a privileged few?

To facilitate safe, sustainable human life in space, we must understand how to effectively feed crew members for extended-duration missions. In this document, we refer to the current model for crew sustainment as the **Terrestrial Sustainment Model (TSM)**. According to TSM, crew members are sustained *primarily* by calorically and nutritionally dense food, often dehydrated, prepared on Earth and transported by the crew. Optionally, when permissible, extended missions are sustained by continuous support from the surface. This is summarized in Figure 1.1.

For the sustainable development of humankind in space, it is clear that *contemporary space operations must abandon TSM* in favor of the **Extraterrestrial Sustainment Model (ETSM)** of crewed space operations. In ETSM, crew members are sustained *primarily* by agricultural operations maintained by the crew themselves. These agricultural operations provide abundant, calorically and nutritionally dense foods, such as cereal grains and vegetables, to crew members. A space mission exercising ETSM will not require continuous support from the surface, allowing for extended crewed missions into deep space. This transition is summarized in Figure 1.2.

Contemporary space technology is not prepared to support a transition to ETSM, as

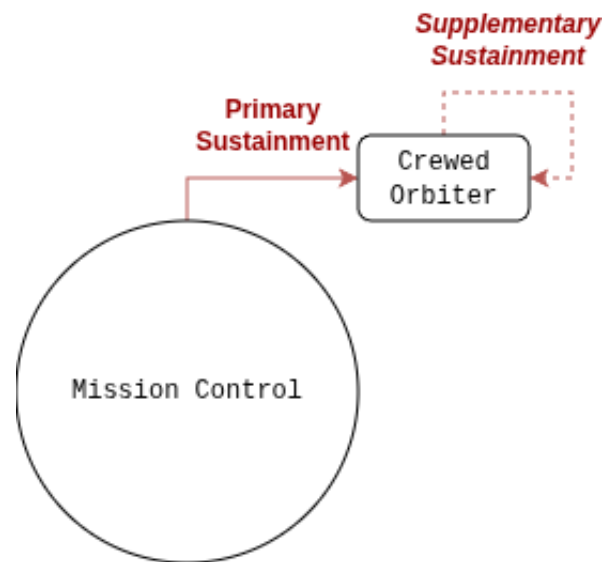


Figure 1.1: Terrestrial Sustainment Model

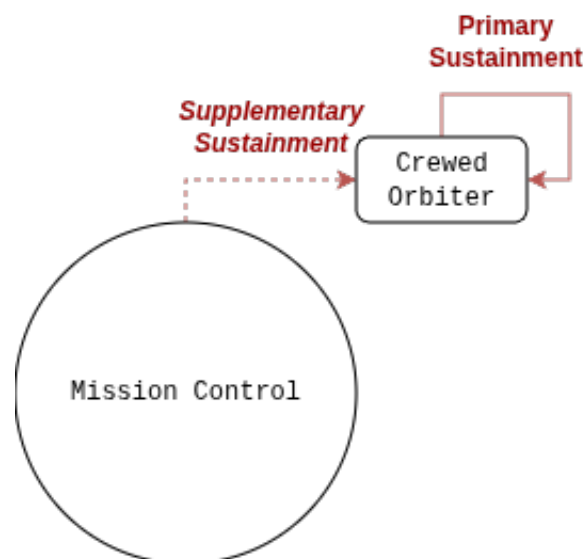


Figure 1.2: Extraterrestrial Sustainment Model

evidenced by the continued need to rely on TSM for crewed mission sustainment. Therefore, to develop the needed technology to transition space operations to ETSM, we propose the **Strategically Positioned Rotating Orbital station to facilitate Understanding and Transition of space agriculture (SPROUT)** mission. This document enumerates the objectives, requirements, subsystems, and impact considerations of the proposed **SPROUT** mission, providing a conceptual framework for future missions to improve space agriculture technology in coming centuries.

1.1.2 Proposed Mission Patch

The proposed mission patch for **SPROUT** is depicted on the cover page of this document and in Figure 1.3. The patch depicts the a silhouette of the Earth (blue) with a stylized young plant (green) superimposed. The plant is oriented at a 45 degree angle and evokes an inclined orbital trajectory, bringing to mind the mission's primary objective to develop sustainable methods of agriculture in space. The circular silhouette of the Earth further suggests the unity of humankind, our noble goal to explore the cosmos, and the **SPROUT** mission's place within that endeavor. Finally, the young plant is divided into three components, representing the intersection of scientific inquiry, skillful engineering, and prudent policy that advances sustainable space development and ensures it is beneficial for all people.



Figure 1.3: Proposed **SPROUT** Mission Patch

1.1.3 Objectives

The **SPROUT** mission has 5 objectives, enumerated below.

OBJ-1 Validate an agricultural environment sustainment system acceptable for growing sufficient calorically and nutritionally dense crops to sustain a six-person crew.

OBJ-2 Identify up to 5 crops optimal for space agriculture to enable extraterrestrial sustenance of crewed missions.

OBJ-3 Determine the optimal growing environment for the crops identified in **OBJ-1**.

OBJ-4 Ensure the health of all crew members and equipment, returning them and their tools to the surface safely at the conclusion of their mission.

OBJ-5 Decommission the **SPROUT** orbiter at the end of its mission using a sustainable EOL plan.

1.2 System Overview

The **SPROUT** system is composed of a crewed **SPROUT orbiter** in LEO and a **SPROUT mission control** station located on the Earth's surface. The crewed orbiter is maintained continuously by a nominal crew of six members: one mission commander (MCDR), one flight surgeon (FLSG), two flight engineers (ENGR), and two scientists (SCIT). The orbiter crew will accomplish all primary mission objectives. The mission control station will continuously monitor the health of the mission crew, oversee scientific experiments conducted on the orbiter, and provide guidance.

The **SPROUT** orbiter consists of four toroids rotating about a central hub. The central hub contains crew quarters and essential systems, and the rotating toroids contain planters for agricultural experimentation. Alternating toroids rotate in opposing directions to minimize excess angular momentum. Each toroid will have a variable rotation rate, allowing their artificial gravity to vary independently and continuously between zero and one multiple of Earth gravity (0.0 m/s^2 to 9.81 m/s^2). The central hub has a cabin pressure of 14.7 psia (11.5 psia N_2 , 3.2 psia O_2), and the four toroids have variable cabin pressure and atmospheric composition. The crew will be sustained by continuous support from the surface as they conduct agricultural experiments.

The **SPROUT** mission control station consists of necessary communication equipment, command and data handling systems, and an unspecified number of Mission Control Operators (MCOP). The MCOPs are responsible for supervising the **SPROUT** crew and their orbiter, ensuring crew health, and directing all scientific experiments.

Chapter 2

System Description

This chapter describes the L1 requirements and subsystems of the **SPROUT** mission. The primary goal of this chapter is to establish requirements flow from mission objectives to subsystem L2 requirements and components.

2.1 System Requirements

Motivated by the **SPROUT** mission objectives, as specified in Section 1.1.3, we define 8 L1 requirements for the **SPROUT** mission. Following each requirement, we list the objectives to which it is pursuant.

REQ-1 The **SPROUT** system shall ensure a safe environment for the crew and their equipment throughout the duration of the mission. (**OBJ-4**)

REQ-2 The **SPROUT** system shall provide a space habitat acceptable to the cultivation of calorically and nutritionally dense agricultural products, capable of sustaining a crew of six. (**OBJ-1, OBJ-2, OBJ-3**)

REQ-3 The **SPROUT** system shall maintain the orbiter's planned orbital trajectory and a stable attitude throughout the duration of the mission. (**OBJ-4**)

REQ-4 The **SPROUT** system shall cultivate selected agricultural products in selected growing conditions to determine their efficacy for extraterrestrial sustainment. (**OBJ-1, OBJ-3**)

REQ-5 The **SPROUT** system shall determine the caloric density, nutritional density, and feasibility of selected agricultural products cultivated in various growing conditions. (**OBJ-3, OBJ-4**)

REQ-6 The **SPROUT** system shall identify up to 5 crops capable of sustaining the **SPROUT** crew and showing potential for future large-scale cultivation in LEO. (**OBJ-2**)

REQ-7 The **SPROUT** system shall find the optimal environmental parameters for maximum caloric density, nutritional density, and sustainability for the crops identified in **REQ-6**. (**OBJ-3**)

REQ-8 The **SPROUT** system shall decommission according to its specified EOL plan, minimizing adverse impact on both the surface and orbital environments. (**OBJ-5**)

2.2 Subsystems

This document identifies 9 key subsystems to support the **SPROUT** mission, and provides an overview of the L2 requirements and components corresponding to each subsystem.

1. **Structures (STR)**. Contains mechanical elements necessary for the structural integrity of the **SPROUT** orbiter.
2. **Power (PWR)**. Provides continuous electrical power to the **SPROUT** orbiter.
3. **Thermal (THM)**. Regulates the flow of heat and its effect on the spacecraft.
4. **Command and Data Handling (CDH)**. Controls information flow across the **SPROUT** system. Analyzes telemetry data.
5. **Guidance, Navigation, and Control (GNC)**. Determines optimal trajectories and develops guidance for the **SPROUT** orbiter.
6. **Attitude Determination and Control (ADC)**. Maintains a stable spacecraft attitude.
7. **Communication (COM)**. Provides data link to mission control on the surface.
8. **Environmental Control and Life Support (ELS)**. Supports mission crew and creates atmospheric and thermal environment amenable to scientific research.
9. **Science (SCI)**. Collects scientific data to support the **SPROUT** mission.

Chapter 3

Structures

The Structures subsystem (STR) provides structural integrity to the **SPROUT** orbiter. It protects the pressurized environment for both crew and agriculture and maintains both the central hub and the toroidal habitats.

The central hub is a cylindrical pressure vessel with an inner radius of 2 m, an outer radius of about 2.5 m, and a length of 59 m. The hub is pressurized to sea level atmospheric conditions and contains crew quarters, essential ELS systems, and CDH systems. The **SPROUT** orbiter also has four independently rotating toroidal habitats. Each habitat has a major radius of 20 m and a minor radius of 2 m. The rotation rate of each habitat can independently vary between 0.0 rad/s and 0.7 rad/s (providing $g_0 = 9.81 \text{ m/s}^2$ of artificial gravity). Appendix B provides a first-order structural analysis outlining the design of these toroidal habitats and demonstrating their physical feasibility.

3.1 Subsystem Requirements

STR has 3 primary L2 requirements, enumerated below.

REQ-STR-1 STR shall provide a pressurized central hub, capable of withstanding up to 50 pressurization cycles between 14.7 psia and vacuum, to a safety factor of 1.25. (**REQ-1**)

REQ-STR-2 STR shall provide four pressurized toroidal habitats, capable of withstanding up to 1000 loading cycles between zero and 9.81 m/s^2 of centrifugal acceleration, as well as 1000 pressurization cycles between 14.7 psia and vacuum, to a safety factor of 1.25. (**REQ-1**, **REQ-2**)

REQ-STR-3 STR shall maintain the structural integrity of the **SPROUT** orbiter throughout launch, orbital insertion, nominal operations, and EOL. (**REQ-1**, **REQ-2**)

3.2 Key Components

The primary structure of the **SPROUT** orbiter is provided by the pressurized central hub and its four independently toroidal habitats. Each toroid is supported along its circumference by a total of eight structural supports. Aluminum 2219 is selected as the primary structural material over composite materials (such as carbon fiber) for its high stiffness and strength, low density, and superior fatigue properties. Appendix B contains a first-order structural analysis of the **SPROUT** orbiter, validating its physical feasibility.

Chapter 4

Power

The Power subsystem (PWR) provides electrical power to all subsystems of the **SPROUT** system. PWR also monitors power usage by each subsystem and integrates hardware watchdogs to control overcurrent events.

4.1 Subsystem Requirements

PWR has 3 primary L2 requirements, enumerated below.

REQ-PWR-1 PWR shall provide at minimum 100 kW of electrical power to the **SPROUT** orbiter. (**REQ-1**, **REQ-2**)

REQ-PWR-2 PWR shall monitor and record power consumption by each subsystem. (**REQ-1**)

REQ-PWR-3 PWR shall prevent overcurrent events using hardware watchdogs. (**REQ-1**)

4.2 Key Components

PWR shall provide necessary electrical power to the **SPROUT** orbiter by means of a radioisotope thermoelectric generator (RTG). Electrical power from the RTG will be provided to every subsystem requiring it. THM is responsible for managing excess thermal energy produced by PWR.

Chapter 5

Thermal

The Thermal subsystem (THM) performs continuous thermal monitoring and management aboard the **SPROUT** orbiter.

5.1 Subsystem Requirements

THM has 3 primary L2 requirements, enumerated below.

REQ-THM-1 THM shall maintain the central hub atmospheric temperature between 293.15 K and 298.15 K. (**REQ-1**)

REQ-THM-2 THM shall maintain the toroidal habitats' atmospheric temperature between 293.15 K and 298.15 K. (**REQ-1**, **REQ-2**)

REQ-THM-3 THM shall reject excess heat generated from all power and information processing systems, including all elements of PWR and CDH. (**REQ-1**)

5.2 Key Components

THM consists of internal atmospheric thermostats, atmospheric heaters, external radiators, and a network of pipes containing a low-GWP working fluid. The thermostats and heaters provide closed-loop feedback for regulating the atmospheric temperature of the pressurized cabins, and the external radiators and low-GWP refrigerant reject excess heat from PWR and CDH.

Chapter 6

Command and Data Handling

The Command and Data Handling subsystem (CDH) is responsible for controlling information flow across all subsystems of **SPROUT**. It provides the central computational infrastructure for the **SPROUT** system, analyzes telemetry data, supervises communications, and supports attitude control and navigation through ADC and GNC.

6.1 Subsystem Requirements

CDH has 4 primary L2 requirements, enumerated below.

REQ-CDH-1 CDH shall monitor information flow to and from all subsystems of **SPROUT**.
(REQ-1, REQ-3)

REQ-CDH-2 CDH shall analyze navigation data from GNC and attitude data from ADC.
(REQ-3)

REQ-CDH-3 CDH shall demonstrate robustness to high TID due to radiative effects.
(REQ-1, REQ-3)

REQ-CDH-4 CDH shall demonstrate robustness to SEU disruptions through the use of redundant computation. (REQ-1, REQ-3)

6.2 Key Components

CDH shall be composed of three radiation-hardened processors surrounded by an appropriate radiation shield. Three processors are used to ensure redundancy and robustness to SEUs. Low power application-specific hardware will compute memory checksums at regular intervals. CDH will also use an application-specific voting module to compare outputs across computations.

Chapter 7

Guidance, Navigation, and Control

The Guidance, Navigation, and Control subsystem (GNC) supervises the orbital trajectory of the **SPROUT** orbiter and executes guidance commands received from mission control on the surface via COM.

7.1 Subsystem Requirements

GNC has 2 primary L2 requirements, enumerated below.

REQ-GNC-1 GNC shall continuously monitor the **SPROUT** orbiter's trajectory and report information to MCOPs via COM. (**REQ-3**)

REQ-GNC-2 GNC shall execute trajectory update commands in coordination with ADC upon instruction from CDH or MCOP. (**REQ-3**)

7.2 Key Components

GNC consists purely of necessary trajectory estimation and control systems. A combination of mission-ready software and, when necessary and appropriate, low power application-specific hardware will be used for numerical integration and trajectory optimization.

Chapter 8

Attitude Determination and Control

The Attitude Determination and Control subsystem (ADC) continuously estimates the attitude of the **SPROUT** orbiter and maintains a stable attitude. ADC uses information from GNC to determine spacecraft trajectory and communicates with mission control via COM.

8.1 Subsystem Requirements

ADC has 3 primary L2 requirements, enumerated below.

REQ-ADC-1 ADC shall continuously estimate the attitude and attitude rate of the **SPROUT** orbiter. (**REQ-3**)

REQ-ADC-2 ADC shall maintain a stable attitude at the setpoint provided by the mission program. (**REQ-3**)

REQ-ADC-3 ADC shall communicate spacecraft attitude estimates to GNC at regular intervals to ensure accurate trajectory planning. (**REQ-3**)

REQ-ADC-4 ADC shall provide reaction torque necessary to counter the effects of unbalanced toroidal rotation.

8.2 Key Components

ADC will be composed of at least three star trackers, necessary computation hardware, the reaction control system (RCS), and a flywheel. The star trackers will identify star patterns to estimate attitude, and the RCS will maintain a stable attitude via closed-loop feedback control. The flywheel will have sufficient inertia to maintain stable roll without effort from the RCS; though alternating toroids rotate in opposite directions to minimize excess angular momentum, a flywheel is still required in cases of imbalanced rotations.

Chapter 9

Communication

The Communication subsystem (COM) maintains radio communication with mission control. COM works with CDH to maintain information flow and integrates watchdogs to restore communication in loss of signal (LOS) events.

9.1 Subsystem Requirements

COM has 3 primary L2 requirements, enumerated below.

REQ-COM-1 COM shall maintain constant communication with mission control on the surface using radio-frequency transmission. (**REQ-1, REQ-3**)

REQ-COM-2 COM shall provide antenna pointing requirements to ADC to ensure consistent downlink. (**REQ-1, REQ-3**)

REQ-COM-3 COM shall communicate with CDH in LOS events to prioritize attitude correction and appropriate power to COM. (**REQ-1, REQ-3**)

9.2 Key Components

COM consists primarily of a radio transceiver, a radio antenna, and necessary supporting hardware. The radio antenna must remain pointed towards mission control receivers, both in orbit and on the surface. COM provides pointing requirements to ADC to ensure appropriate spacecraft attitude.

Chapter 10

Environmental Control and Life Support

The Environmental Control and Life Support subsystem (ELS) is responsible for maintaining a stable atmospheric and thermal environment for the **SPROUT** crew, equipment, and experiments.

10.1 Subsystem Requirements

ELS has 4 primary L2 requirements, enumerated below. Metrics provided are based on the National Aeronautics and Space Administration's environmental control and life support guidelines [1].

REQ-ELS-1 ELS shall maintain a consistent atmospheric environment of 14.7 psia in crew quarters, composed of 11.5 psia N_2 and 3.2 psia O_2 . (**REQ-1**)

REQ-ELS-2 ELS shall generate at least 10 kg O_2 daily, supplying 5 kg to crew and maintaining an emergency supply of 300 kg. (**REQ-1**)

REQ-ELS-3 ELS shall scrub CO_2 from crew quarters continuously and convert it to O_2 or distribute it to agricultural experiments. (**REQ-1, REQ-2, REQ-4**)

REQ-ELS-4 ELS shall provide at least 25 L of potable H_2O daily to crew members, reclaiming wastewater when feasible. (**REQ-1, REQ-2**)

REQ-ELS-5 ELS shall maintain a consistent atmospheric environment in toroidal growing environments as specified by mission SCIT. (**REQ-2, REQ-4**)

10.2 Key Components

ELS is composed primarily of the following components.

1. **Atmospheric regulator.** Maintains appropriate partial pressures of N_2 and O_2 in all pressurized cabins, and continuously monitors and removes particulates and VOCs.
2. **Oxygenator.** Reclaims O_2 from CO_2 , provides the atmospheric regulator with sufficient gaseous O_2 , and stores excess O_2 in pressurized tanks.
3. **Water reclaimer.** Scrubs wastewater to obtain potable H_2O .

Chapter 11

Science

The Science subsystem (SCI) provides necessary scientific instruments and analysis tools to the **SPROUT** system. It determines caloric and nutritional densities and sustainabilities for selected crops cultivated on board the **SPROUT** orbiter. SCI distinguishes between two types of agricultural products cultivated on the **SPROUT** orbiter: **cereal grains** and **powerhouse crops**. Cereal grains are cultivated primarily for their high caloric density, providing the basis for a low glycemic index (LGI) diet for the crew; for example, rice and maize are candidate cereal grains. Powerhouse crops are cultivated primarily for their high micronutrient density, providing critical antioxidants, vitamins, and minerals (CAVM) to the crew's diet; for example, spinach and kale are candidate powerhouse crops. Candidate cereal grains, candidate powerhouse crops, and CAVM are enumerated in Section 11.3.

11.1 Subsystem Requirements

SCI has 12 primary L2 requirements, enumerated below.

REQ-SCI-1 SCI shall determine the mass density of bioavailable glucose present in cultivated cereal grains. (**REQ-5**)

REQ-SCI-2 SCI shall determine the glycemic index (GI) of bioavailable glucose present in cultivated cereal grains. (**REQ-5**)

REQ-SCI-3 SCI shall determine the required water per unit mass of bioavailable glucose present in cultivated cereal grains. (**REQ-5**)

REQ-SCI-4 SCI shall determine the required carbon dioxide per unit mass of bioavailable glucose present in cultivated cereal grains. (**REQ-5**)

REQ-SCI-5 SCI shall determine the required fertilizer per unit mass of bioavailable glucose present in cultivated cereal grains. (**REQ-5**)

REQ-SCI-6 SCI shall determine the required energy per unit mass of bioavailable glucose present in cultivated cereal grains. (**REQ-5**)

REQ-SCI-7 SCI shall determine the mass density of bioavailable CAVM present in cultivated powerhouse crops. (**REQ-5**)

REQ-SCI-8 SCI shall determine the required water of bioavailable CAVM present in cultivated powerhouse crops. (**REQ-5**)

REQ-SCI-9 SCI shall determine the required carbon dioxide of bioavailable CAVM present in cultivated powerhouse crops. (**REQ-5**)

REQ-SCI-10 SCI shall determine the required fertilizer of bioavailable CAVM present in cultivated powerhouse crops. (**REQ-5**)

REQ-SCI-11 SCI shall determine the required energy per unit mass of bioavailable CAVM present in cultivated powerhouse crops. (**REQ-5**)

REQ-SCI-12 SCI shall determine the optimal gravitation strength for each candidate cereal grain and powerhouse crop. (**REQ-5**)

11.2 Key Components

SCI is composed of various tools for the cultivation and analysis of the crops cultivated aboard the **SPROUT** orbiter. We decompose these needs into four primary classes of scientific instrumentation.

1. **Cultivation.** Cultivation instruments are used to support the growth and maturation of candidate cereal grains and powerhouse crops. These instruments include various irrigation and lighting systems.
2. **Life Cycle Analysis.** Life cycle analysis instruments are used to monitor the life cycle of candidate crops. These tools will detect and record the growth rate of crops under various conditions, assisting the scientific crew in determining the optimal growing conditions for crops.
3. **Nutrition Analysis.** Nutrition analysis instruments are used to determine the nutritional content of mature crops. Once reaching maturation, crops will be harvested by the scientific crew and analyzed for bioavailable glucose, bioavailable CAVM, and glycemic index.

4. **Transcriptomics.** Transcriptomic instruments will harness the immense power of modern sequencing technologies and large-scale, low-power computation to determine expression of various genes associated with physiological maturation and response to environmental variables.

11.3 Experimental Candidates and CAVM

11.3.1 Candidate Cereal Grains

To efficiently provide crews on deep space missions with LGI diets, the **SPROUT** mission will cultivate and determine the feasibility of cereal grains in a space habitat. The following cereal grains are proposed.

1. Maize
2. Rice
3. Wheat

11.3.2 Candidate Powerhouse Crops

Cereal grains can be cultivated densely and efficiently provide human crews with LGI caloric value. Though necessary, cereal grains are woefully insufficient for a balanced and holistic human diet. Therefore, the **SPROUT** mission proposes the cultivation of powerhouse crops, cultivated primarily for their high density of CAVM.

1. Watercress
2. Carrots
3. Various cultivars of *B. oleracea*, including broccoli, kale, and cauliflower

11.3.3 Critical Antioxidants, Vitamins, and Minerals

The **SPROUT** mission seeks to identify feasible crops that provide highly bioavailable critical antioxidants, vitamins, and minerals (CAVM). This document proposes the following CAVM to be researched by the onboard scientific crew. This list is not meant to be extensive—rather, it is meant to prompt a conversation between nutritionists, food scientists, and biologists around which CAVM are of highest priority to sustain human health in space.

1. Vitamin C
2. Vitamin E
3. Beta-Carotene

4. Iron

5. Selenium

Chapter 12

System Interfaces

This section establishes design structure between the subsystems described in previous chapters. Table 12.1 presents the Design Structure Matrix (DSM) for the **SPROUT** mission. Columns designate systems providing “outputs” and rows designate systems requiring “inputs”. We abide by the following convention.

- *str.* — structural interface
- *elec.* — power electronic interface
- *thm.* — thermal interface
- *info.* — information interface

Table 12.1: Design Structure Matrix (DSM) for **SPROUT**

DSM	STR	PWR	THM	CDH	GNC	ADC	COM	ELS	SCI
STR	—								
PWR	<i>str.</i>	—	<i>elec.</i>	<i>info.</i>					
THM	<i>str.</i>	<i>elec. / thm.</i>	—	<i>info. / thm.</i>				<i>thm.</i>	
CDH	<i>str.</i>	<i>info. / elec.</i>	<i>info.</i>	—	<i>info.</i>	<i>info.</i>	<i>info.</i>	<i>info.</i>	<i>info.</i>
GNC	<i>str.</i>	<i>elec.</i>		<i>info.</i>	—	<i>info.</i>	<i>info.</i>		
ADC	<i>str.</i>	<i>elec.</i>		<i>info.</i>		—			
COM	<i>str.</i>	<i>elec.</i>		<i>info.</i>			—		
ELS	<i>str.</i>	<i>elec.</i>	<i>thm.</i>	<i>info.</i>				—	
SCI	<i>str.</i>	<i>elec.</i>							—

Chapter 13

Impact Considerations

The **SPROUT** mission stands to make a critical impact on the transition of humanity to a spacefaring species by facilitating sustainable space agriculture. Beginning the transition of human crews to ETSM by the year 2040, and transitioning entirely to ETSM by approximately the year 2070, would greatly accelerate ongoing projects in space urbanism.

The **SPROUT** mission, in a special way, is also meant to be a successor to the Aurelia Institute's Space Garden project. The Space Garden project is an "orbiting, autonomous greenhouse" allowing the selection and stewardship of crops viable for cultivation in space [4]. The goals and outcomes of this project are naturally extended into a long-term mission, feasible within a decade of this writing, that could enable complete transition to ETSM for future human crews.

We anticipate many additional positive impacts on the sustainable development of humankind due to the unprecedented research conducted by the **SPROUT** mission; however, we consider further elaboration out of scope for this document.

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Appendix A

Abbreviations

Table A.1 enumerates various abbreviations and nomenclature used throughout this document.

Table A.1: Nomenclature in This Document

Abbreviation	Meaning
ADC	Attitude Determination and Control subsystem
CAVM	Critical Antioxidants, Vitamins, and Minerals
CDH	Command and Data Handling subsystem
COM	Communication subsystem
DSM	Design Structure Matrix
ELS	Environmental control and Life Support subsystem
ENGR	Engineer
EOL	End Of Life
ETSM	Extraterrestrial Sustainment Model
FLSG	Flight Surgeon
GI	Glycemic Index
GNC	Guidance, Navigation, and Control subsystem
GWP	Global Warming Potential
L1	Level 1 requirement
L2	Level 2 requirement
LEO	Low Earth Orbit
LGI	Low Glycemic Index
LOS	Loss Of Signal
MCDR	Mission Commander
MCOP	Mission Control Operator
PWR	Power subsystem
RCS	Reaction Control System
REQ	Requirement
RTG	Radioisotope Thermoelectric Generator
SAND	Simultaneous Analysis And Design
SCI	Science subsystem
SCIT	Scientist
SEU	Single Event Upset
SPROUT	Strategically Positioned Rotating Orbital station to facilitate Understanding and Transition of space agriculture
STR	Structures subsystem
THM	Thermal subsystem
TID	Total Ionizing Dose
TSM	Terrestrial Sustainment Model
VOC	Volatile Organic Compound

Appendix B

Analysis and Design of Rotating Toroids

This section demonstrates the physical feasibility of four counter-rotating toroids used for agricultural experimentation via simultaneous analysis and design (SAND) methods. Throughout, we assume that the toroids and their support structures are manufactured from aluminum 2219, with material properties given by [3]. In this section, we perform a preliminary structural analysis and design of the rotating toroids and their support structures by framing the design problem as an optimization program. Figure B.1 shows a cartoon of the proposed design for a single toroid (not to scale).

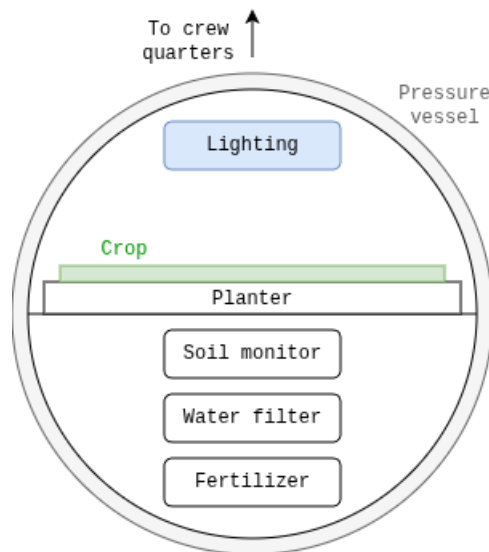


Figure B.1: Proposed Design Cross Section for **SPROUT** Toroid

Nomenclature

A_C	cross-sectional area of toroid core
A_f	minimum required cultivation area
A_r	cross-sectional area of radial support structure
A_x	cross-sectional area of load-bearing elements along toroidal ring
a	strength of artificial gravity, in multiples of g_0
E_{Al}	elastic modulus of aluminum 2219
ϵ_{rr}	radial strain of radial support structure
$\epsilon_{\theta\theta}$	hoop strain of toroid
η	engineering factor of safety
η_s	cultivation area usage efficiency
g_0	acceleration due to Earth's gravity at sea level
γ	maximum permissible gravity differential
N	internal force of radial support structure
n	number of independently rotating toroids
Ω	angular velocity of toroidal rotation
R	major radius of toroid
r	major radial coordinate of toroid
ρ_{Al}	density of aluminum 2219
ρ_C	mean density of toroid core, including atmosphere, equipment, and agriculture
σ_{rr}	radial stress of radial support structure
σ_Y	yield stress of aluminum 2219
$\sigma_{\theta\theta}$	hoop stress of toroid
T	inner minor radius of toroid
θ	circumferential coordinate of toroid
u_r	radial displacement field of radial support structure
$\hat{x}$	unit vector in x direction
$\hat{y}$	unit vector in y direction

B.1 Required Cultivation Area

Traditional Earth-bound agricultural methods typically support one person per acre of arable land ($\approx 4046.86 \text{ m}^2$) [5]. Furthermore, with advances in agricultural technologies, novel methods such as vertical farming can be 10 to 20 times more productive per unit area; in this document, we will use the conservative estimate that internal farming is 15 times more productive than traditional farming [2]. This suggests that, for a crew of six, at least $A_f = 1618.74 \text{ m}^2$ of cultivation area is required on board the **SPROUT** orbiter. Divided among $n = 4$ toroids, this requires a minimum of $A_f/n = 404.69 \text{ m}^2$ of cultivation area per toroid.

We assume throughout this appendix that circular cross sections are desirable to maintain the orbiter's pressure vessel. Therefore, the cultivation area is constrained by

$$\frac{A_f}{n} \leq 4\pi R T \eta_s \quad (\text{B.1})$$

to ensure that sufficient space is available in each toroid to support cultivation.

B.2 Artificial Gravitation

Artificial gravitation is provided in the rotating toroids by the “fictitious” centrifugal force. The strength of artificial gravity, given in multiples of g_0 , is approximately equivalent to

$$a = \frac{\Omega^2 R}{g_0}. \quad (\text{B.2})$$

In (B.2), and throughout this study, we assume that the interior of the toroid is negligible in size compared to the distance R . It is desirable that a may vary between zero and unity; we hypothesize that growing crops in gravitation stronger than that at Earth’s surface will not be advantageous.

B.3 Hoop Stress and Strain in Toroid

The rotation of the toroidal structure will induce a **hoop stress** $\sigma_{\theta\theta}$. An infinitesimal element of the toroid has mass $(A_x \rho_{AI} + A_C \rho_C) R d\theta$ and (in the rotating frame) experiences an acceleration outward at $\Omega^2 R$. Therefore, the force on the section of the toroid between $\theta = 0$ and $\theta = \pi$ is given by

$$\vec{F} = \int_{\theta=0}^{\theta=\pi} (A_x \rho_{AI} + A_C \rho_C) R^2 \Omega^2 (\hat{x} \cos \theta + \hat{y} \sin \theta) d\theta \quad (\text{B.3})$$

$$= (A_x \rho_{AI} + A_C \rho_C) R^2 \Omega^2 \hat{y}. \quad (\text{B.4})$$

This force must be balanced by twice the hoop stress in the toroidal (provided at $\theta = 0$ and $\theta = \pi$). Therefore, the hoop stress in the toroid is

$$\sigma_{\theta\theta} = \frac{R^2 \Omega^2 (A_x \rho_{AI} + A_C \rho_C)}{2A_x}. \quad (\text{B.5})$$

Assuming that the load is entirely carried by the structural elements of the toroid, we can write the **hoop strain** $\epsilon_{\theta\theta}$ as

$$\epsilon_{\theta\theta} = \frac{\sigma_{\theta\theta}}{E_{AI}} = \frac{R^2 \Omega^2 (A_x \rho_{AI} + A_C \rho_C)}{2E_{AI} A_x}. \quad (\text{B.6})$$

We know that this is also equal to $\Delta R/R$ by the definition of hoop strain. This fact will be extremely useful to us shortly.

B.4 Stress and Displacement in Radial Support Structure

We model each radial support structure as a column with cross-sectional area A_r and elastic modulus E_r . An infinitesimal element of the support structure, located at distance r from the center of rotation, experiences a radial force (in the rotating frame) of $\rho_{Al}A_r\Omega^2r$. This force must equal exactly the rate of change of the internal force with respect to the radius:

$$\frac{\partial N}{\partial r} + \rho_{Al}A_r\Omega^2r = 0. \quad (B.7)$$

We solve this differential equation for $N(r)$, obtaining

$$N(r) = -\frac{1}{2}\rho_{Al}A_r\Omega^2r^2 + E_{Al}A_rC_1 \quad (B.8)$$

where C_1 is a unitless constant of integration specified by boundary conditions. From this, we obtain the radial stress σ_{rr}

$$\sigma_{rr}(r) = \frac{N(r)}{A_r} = -\frac{1}{2}\rho_{Al}\Omega^2r^2 + E_{Al}C_1 \quad (B.9)$$

and the radial strain ε_{rr} .

$$\varepsilon_{rr}(r) = \frac{\sigma_{rr}(r)}{E_{Al}} = -\frac{\rho_{Al}\Omega^2}{2E_{Al}}r^2 + C_1 \quad (B.10)$$

Finally, using the relationship $\varepsilon_{rr} = \frac{\partial u_r}{\partial r}$, we compute $u_r(r)$ to be

$$u_r(r) = -\frac{\rho_{Al}\Omega^2}{6E_{Al}}r^3 + C_1r + C_2R \quad (B.11)$$

where C_2 is an additional constant of integration specified by boundary conditions. The applicable boundary conditions are

$$u_r(r = 0) = 0, \text{ and} \quad (B.12)$$

$$u_r(r = R) = R\varepsilon_{\theta\theta} = \frac{R^3\Omega^2(A_x\rho_{Al} + A_C\rho_C)}{2E_{Al}A_x}. \quad (B.13)$$

The first is applicable because the support structure must remain connected to the driving device at the central hub, and the second is applicable because the support structure must remain connected to the toroid, which is deformed according to the hoop strain given in (B.6). Applying these in order, we find

$$u_r(r = 0) = 0 \quad (\text{B.14})$$

$$\Rightarrow C_2 = 0, \text{ and} \quad (\text{B.15})$$

$$u_r(r = R) = \frac{R^3 \Omega^2 (A_x \rho_{Al} + A_C \rho_C)}{2 E_{Al} A_x} \quad (\text{B.16})$$

$$\Rightarrow C_1 = \frac{R^2 \Omega^2}{2 E_{Al}} \left(\frac{4}{3} \rho_{Al} + \frac{A_C}{A_x} \rho_C \right). \quad (\text{B.17})$$

$$(\text{B.18})$$

This gives us final expressions for $u_r(r)$, $\varepsilon_{rr}(r)$, and $\sigma_{rr}(r)$.

$$u_r(r) = -\frac{\rho_{Al} \Omega^2}{6 E_{Al}} r^3 + \frac{R^2 \Omega^2}{2 E_{Al}} \left(\frac{4}{3} \rho_{Al} + \frac{A_C}{A_x} \rho_C \right) r \quad (\text{B.19})$$

$$\therefore \varepsilon_{rr}(r) = -\frac{\rho_{Al} \Omega^2}{2 E_{Al}} r^2 + \frac{R^2 \Omega^2}{2 E_{Al}} \left(\frac{4}{3} \rho_{Al} + \frac{A_C}{A_x} \rho_C \right) \quad (\text{B.20})$$

$$\therefore \sigma_{rr}(r) = -\frac{\rho_{Al} \Omega^2}{2} r^2 + \frac{R^2 \Omega^2}{2} \left(\frac{4}{3} \rho_{Al} + \frac{A_C}{A_x} \rho_C \right) \quad (\text{B.21})$$

B.5 Differential Gravitation

Our final consideration is that of **differential gravitation**. In this artificial gravity environment, objects closer to the “roof” of the toroidal core will experience a weaker acceleration (in the rotating frame) than objects closer to the “floor” of the core. We seek to keep the strength of artificial gravity near the “roof” to be at least $1 - \gamma$ times that at the “floor”. Formally,

$$\frac{\Omega^2(R - T)}{\Omega^2 R} \geq 1 - \gamma \quad (\text{B.22})$$

$$\therefore 1 - \frac{T}{R} \geq 1 - \gamma \quad (\text{B.23})$$

$$\therefore \frac{T}{R} \leq \gamma \quad (\text{B.24})$$

B.6 Optimization Objective and Constraints

As a proxy for construction cost and maintenance difficulty, we select to minimize the surface area of a single toroid. However, we are constrained by the yield stress of the toroid and the yield stress of a structural support element. The maximum stress will be experienced when the artificial gravitation is at maximum ($a = 1$) so we impose this constraint.

$$\Omega^2 R = g_0 \quad (\text{B.25})$$

We also impose the constraint that the maximum radial stress multiplied by η must not exceed σ_Y , the yield stress of aluminum 2219. The maximum radial stress is given at $r = 0$, which maximizes (B.21).

$$\sigma_{rr}(r = 0) = \frac{R^2\Omega^2}{2} \left(\frac{4}{3}\rho_{Al} + \frac{A_C}{A_x}\rho_C \right) \leq \frac{\sigma_Y}{\eta} \quad (\text{B.26})$$

Finally, we impose the constraint that the hoop stress $\sigma_{\theta\theta}$ multiplied by η must not exceed the yield stress of aluminum 2219.

$$\sigma_{\theta\theta} = \frac{R^2\Omega^2 (A_x\rho_{Al} + A_C\rho_C)}{2A_x} \leq \frac{\sigma_Y}{\eta} \quad (\text{B.27})$$

It is interesting to note that the satisfaction of (B.26) guarantees the satisfaction of (B.27), because aluminum has never been observed to have non-positive density (i.e., $\rho_{Al} > 0$). We demonstrate this as follows.

$$\frac{R^2\Omega^2}{2} \left(\frac{4}{3}\rho_{Al} + \frac{A_C}{A_x}\rho_C \right) > \frac{R^2\Omega^2 (A_x\rho_{Al} + A_C\rho_C)}{2A_x} \quad (\text{B.28})$$

$$\Leftrightarrow \frac{4}{3}\rho_{Al} + \frac{A_C}{A_x}\rho_C > \frac{A_x\rho_{Al} + A_C\rho_C}{A_x} \quad (\text{B.29})$$

$$\Leftrightarrow \frac{4}{3}\rho_{Al} + \frac{A_C}{A_x}\rho_C > \rho_{Al} + \frac{A_C}{A_x}\rho_C \quad (\text{B.30})$$

$$\Leftrightarrow \rho_{Al} > 0, \text{ obviously true} \quad (\text{B.31})$$

Therefore, it is safe to discard the constraint in (B.27) when optimizing, because (B.26) will guarantee its satisfaction. Including all previous constraints, including the constraints of (B.24) and (B.1), we are now ready to express our optimization problem as

$$\min_{R, \Omega, T} 4\pi^2 RT \quad (\text{B.32})$$

$$\text{subject to } \Omega^2 R = g_0 \quad (\text{B.33})$$

$$\frac{R^2\Omega^2}{2} \left(\frac{4}{3}\rho_{Al} + \frac{A_C}{A_x}\rho_C \right) \leq \frac{\sigma_Y}{\eta} \quad (\text{B.34})$$

$$T \leq \gamma R \quad (\text{B.35})$$

$$\frac{A_f}{n} \leq 4\pi RT \eta_s. \quad (\text{B.36})$$

Performing a necessary substitution, using $A_C = \pi T^2$, and simplifying, we obtain the quadratically constrained quadratic program

$$\min_{R, \Omega, T} RT \quad (\text{B.37})$$

$$\text{subject to } \frac{Rg_0}{2} \left(\frac{4}{3}\rho_{\text{Al}} + \frac{\pi T^2}{A_x}\rho_{\text{C}} \right) \leq \frac{\sigma_Y}{\eta} \quad (\text{B.38})$$

$$T \leq \gamma R \quad (\text{B.39})$$

$$\frac{A_f}{n} \leq 4\pi RT\eta_s \quad (\text{B.40})$$

$$R \geq 0 \quad (\text{B.41})$$

$$T \geq 0 \quad (\text{B.42})$$

in which the final two constraints are applied to avoid physically impossible constructions. We rearrange this optimization program into a more familiar form to solve.

$$\min_{R, T} RT \quad (\text{B.43})$$

$$\text{subject to } R \left(\frac{2}{3}\rho_{\text{Al}} + \frac{\pi}{2A_x}\rho_{\text{C}}T^2 \right) \leq \frac{\sigma_Y}{\eta g_0} \quad (\text{B.44})$$

$$T \leq \gamma R \quad (\text{B.45})$$

$$RT \geq \frac{A_f}{4\pi\eta_s n} \quad (\text{B.46})$$

$$R \geq 0 \quad (\text{B.47})$$

$$T \geq 0 \quad (\text{B.48})$$

Unfortunately, the optimization program given by (B.43) is nonconvex. However, given the values

$$\begin{aligned} \rho_{\text{Al}} &= 2840 \text{ kg/m}^3 \\ \rho_{\text{C}} &= 1000 \text{ kg/m}^3 \quad (\text{first-order estimate}) \\ A_x &= 0.2 \text{ m}^2 \quad (\text{first-order estimate}) \\ \sigma_Y &= 103 \text{ MPa} \quad (\text{fatigue stress is used to mitigate cyclic loading}) \\ \eta &= 1.25 \\ \gamma &= 0.1 \\ A_f &= 1618.74 \text{ m}^2 \\ \eta_s &= 0.8 \\ n &= 4 \end{aligned}$$

it can be shown by inspection that a global minimizer is approximately

$$R = 20 \text{ m} \quad (\text{B.49})$$

$$T = 2 \text{ m} \quad (\text{B.50})$$

$$\Omega = 0.7 \text{ rad/s.} \quad (\text{B.51})$$

Thus we elect to construct the **SPROUT** orbiter according to these parameters.